

Financial News Sentiment vs. NIFTY50 — Project Summary

Anmol Krishna | Roll No. 24051240 | KIIT VISTA Data Science Bootcamp | K-Explore Elective: Web Designing

DATA

3,349 raw headlines (Feb–Aug 2025) → 3,348 after removing one exact duplicate → 173 unique dates, after dropping one malformed row where article text had spilled into the Date column → inner-joined with 142 NIFTY50 trading days to give **132 usable rows**. Zero nulls and zero failed date parses on either side; dropped days are weekends, holidays and the windows outside the news feed's coverage.

KEY EDA FINDING

The rule-based score is strongly positive-skewed: mean **+18.17**, median +13, only 26 of 132 days at or below zero — yet the market closed higher on just 65 days (49%). Correlation with same-day movement is **0.094**, effectively zero. Three mechanical causes, found by inspecting the scorer rather than trusting it: `.count()` matches substrings rather than words; stem families double-count, since 'gain' and 'gains' are both listed; and the unbounded score scales with headline volume, which ranges from 2 to 75 per day.

PART A — SAME-DAY (RULE-BASED)

Accuracy **59.85%** against a majority-class baseline of **50.76%**. The edge is concentrated, not general: the rule calls UP on 106 of 132 days and is right only 56% of the time when it does, versus 77% on the 26 days it calls DOWN. This is **co-movement, not prediction** — sentiment and target come from the same trading day, so it could not be traded.

PART B — NEXT-DAY (MACHINE LEARNING)

Target shifted forward one trading day; 125 rows, split chronologically into 100 training and 25 test days (16 Jul – 20 Aug 2025). No shuffling, no leakage.

Model	Test accuracy	vs baseline
Majority-class baseline	56.00%	—
Random Forest (best model)	52.00%	-4.00
Logistic Regression	44.00%	-12.00
XGBoost	44.00%	-12.00
Tuned Random Forest (CV 51.25%)	44.00%	-12.00

No model beats the baseline. Walk-forward validation over five independent windows agrees: Random Forest 51.00%, Logistic Regression 52.00%, against per-fold baselines averaging 54.00%. GridSearchCV (TimeSeriesSplit) selected the *most constrained* configuration in the grid, and tuning made test accuracy worse (52% to 44%) — both are what happens when validation performance is noise. Feature importances split evenly (sentiment 0.511, price/volume 0.489), but raw sentiment_score ranks seventh of eight while smoothed sentiment_ma7 ranks second; these describe what the model split on, not what predicts the market.

A naive long-only backtest returned **Rs 0.9923** versus **Rs 0.9936** buy-and-hold on Rs 1. Both lost money in a declining window, and the strategy's mid-period lead came from sitting in cash, not timing. Costs, slippage and taxes are excluded; the model predicts *intraday* direction while the backtest realises *close-to-close* returns, and the final day is undefined, so 24 transitions are simulated. **Illustrative, not a trading signal.**

PART C — ROBUSTNESS OF THE SENTIMENT MEASURE (SELF-DIRECTED)

Auditing every term against a word-boundary regex exposed **841 phantom positive** and **1,255 phantom negative** matches. The term 'ban' alone accounted for 378, with **zero** occurrences as an actual word — every hit inside "bank" or "banking", scored negative, in a corpus dominated by banking coverage.

Sentiment variant (same-day, 132 days)	Correlation	Accuracy	UP calls
Majority-class baseline	—	50.76%	—
Original (substring)	0.0943	59.85%	106
<code>''.join()</code> stretch goal	0.0943	59.85%	106
Regex word-bounded	0.0429	53.03%	119
Regex + normalised	0.1438	53.03%	119
VADER	-0.1170	46.97%	129
TextBlob	0.0132	50.00%	95

The `''.join()` stretch goal produced identical scores on all 132 days, because `.count()` matches substrings and the punctuation `str(list)` introduces never falls inside a vocabulary term.

Correcting the scorer lowered same-day accuracy to 53.03% — the opposite of the expected result. Word-boundary matching raised the minimum score from -24 to -12 and cut DOWN calls from 26 days to 13; since Part A's edge lived entirely in those strongly-negative days, the phantom matches were *manufacturing* that negativity, and removing them removed the edge. Normalising by headline count gave the strongest correlation of any variant (**0.144**). VADER scored **below chance**, consistent with general-purpose tools mis-reading financial vocabulary. Re-running Part B on each variant changed nothing: none beat the 56% baseline, and a price-only feature set did no better — the constraint is sample size, not the sentiment measure.

CONCLUSION

Sentiment **co-moves** with same-day NIFTY50 direction (59.85% vs 50.76%) but carries **no next-day forecasting power**, and the same-day edge itself substantially dissolves once a scoring artefact is corrected. Consistent with public news being absorbed into prices almost instantly; reported as found rather than as hoped for.

WHAT I WOULD DO DIFFERENTLY WITH MORE TIME

The binding constraint is data volume: 132 trading days cannot separate a weak signal from noise, and a 25-day test set makes a 12-point accuracy gap roughly three days of difference. I would extend both series to several years, then benchmark the corrected scorer against FinBERT, which is trained on financial text. I would also align the prediction target with the traded return, and tune the decision threshold rather than fixing it at zero, since the normalised score's higher correlation was not exploitable by a > 0 rule. Most promising, intraday timestamps would separate headlines published before the close from those after — the likeliest place for genuine signal to hide.